“Weather Sensitivity and Bike-Sharing Demand: Evidence from Weekday-Weekend Interactions in Seoul”

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Abstract

Weather conditions are a key driver of demand volatility in urban bike-sharing systems; yet existing research often treats users as homogeneous groups based on subscription types, overlooking behavioural differences in trip purpose. This study examined whether weather sensitivity varies between weekdays and weekends, using hourly rental data from Seoul's public bike-sharing system. Applying multiple linear regression with interaction terms on 1,447 observations from May-June 2018, we examined how temperature, humidity and rainfall affect bike rental demand across day types.

Results indicated that temperature and humidity exhibit statistically significant differential effects: weekend demand increases more rapidly with rising temperatures, while humidity's negative effect on weekday bike rentals is reduced on weekends. In contrast, rainfall effects, despite appearing different in exploratory analysis, did not vary significantly between weekdays and weekends in formal hypothesis testing.

These findings suggest that day-type segmentation captures behavioural regime shifts more effectively than subscription-based approaches, with practical implications for dynamic pricing, demand forecasting, and fleet management in bike-sharing and micromobility operations.

Introduction

As an avid Lime user in London, I've noticed that I always check weather forecasts before weekend rides but rarely before weekday university commutes. This suggests that the purpose of my trip moderates my sensitivity to weather when choosing to bike-share. These behaviours can have significant operational implications: major bike-sharing operators, such as Lime, Lyft, and Citi Bikes, face volatile demand driven by weather conditions. However, current research informing business decisions treats users as homogeneous groups categorised by subscription models, an approach that overlooks decision-making based on utilitarian or recreational usage. With the number of global bike-sharing users expected to reach 1.2bn by 2030 (Statista, 2025), understanding whether weather sensitivity differs by trip purpose could inform dynamic pricing strategies (potential 5-10% revenue increase), optimised fleet management (reduced operational costs), and improved demand forecasting accuracy. This study investigates the regime shifts in weather sensitivity between weekdays and weekends using Seoul's public bike-sharing data, using day type as an operational proxy for trip purpose. Weekdays are treated as predominantly utilitarian (commuting-oriented) travel and weekends as recreational (leisure-oriented) usage.

Recent research shows that bike-sharing usage exhibits distinct temporal patterns, with weekday demand characterised by bimodal commute peaks between 07:00 to 09:00 and 17:00 to 19:00. Meanwhile, weekend demand shows flatter distributions indicative of recreational activities (Kinoshita, Bando and Sayama, 2024). According to V.E. and Cho (2020), weather conditions, especially temperature, also significantly influence bike-sharing demand. Recent studies in Seoul found that recreational users – identified through one-day passes – demonstrate greater weather sensitivity than commuters with long-term passes (Kim, 2023). However, subscription type conflates user identity with trip purpose: Kim notes that long-term pass holders also engage in recreational cycling, especially on weekends, hiding behavioural segmentation. No prior study has examined whether a split between weekdays and weekends provides a clearer test of utilitarian versus recreational weather sensitivity. Analysing day-specific usage factors can inform operational strategies like weekend surge pricing, managing fleets and maintenance during downtimes, and forecasting demand given weather data – business implications available only through granular, day-type usage breakdowns.

This study addresses the research question: **Does the effect of weather conditions on bike rental demand differ between weekdays and weekends in Seoul's public bike-sharing system?** We test whether day type (weekday/weekend) reveals clearer behavioural regime shifts by examining three weather variables:

temperature, humidity, and rainfall. Using multiple linear regression on hourly rental data (n = 1,464) from May-June 2018, we test the null hypothesis (H_0) that **weather variables affect bike rental demand equally across weekdays and weekends**, and the alternative hypothesis (H_1) that **at least one weather variable exhibits a significantly different effect on bike rental demand between weekdays and weekends.** Our findings contribute by showing that day-type segmentation captures behavioural regime shifts more cleanly than subscription-type segmentation. If weather sensitivity varies by day type, operators can implement targeted interventions – pricing and fleet rebalancing – aligned with behavioural differences between utilitarian and recreational trip purposes.

Data & Methodologies

Data was sourced from a modified version of the 'Seoul Bike Sharing Demand' dataset from the UC Irvine Machine Learning Repository. The dataset comprises 1,464 observations spanning May 1, 2018, to June 30, 2018, with hourly bike rental counts and associated weather conditions.

Python was used in this study for data validation, processing, visualisation and statistical modelling. Data manipulation was performed using the numpy and pandas libraries, while matplotlib and seaborn provided exploratory visualisations. Linear and multiple regression modelling, along with hypothesis testing, were conducted using scipy and statsmodels.

Data processing involved retaining and renaming key variables: '**Date' (renamed 'date'), 'Rented Bike Count '('rentals'), 'Hour' ('hour'), 'Temperature (°C)' ('temp'), 'Humidity (%)' ('humidity'),** and **'Rainfall (mm)' ('rainfall')**. Seventeen observations with 0% humidity were removed as data quality anomalies. A binary variable, '**is_weekend**', was created to distinguish weekend from weekday observations. The final processed dataset contained 1,447 observations: 1,042 weekday and 405 weekend records.

A log-transformation of rainfall, $ln(1 + x)$, where x is the original rainfall value in mm, was applied to correct right-skewness, creating **'rainfall_log'**. The original rainfall distribution is severely right-skewed, with most values at zero and outliers present. Applying a natural logarithmic transformation compresses extreme values while expanding the lower range. This better captures the non-linear relationship between rainfall and bike rentals, where initial rainfall has a stronger deterrent effect that gradually increases at higher levels. As shown in Figure 1, log-transformed rainfall shows improved linearity; each 1-unit increase corresponds to ~550 fewer bike rentals.

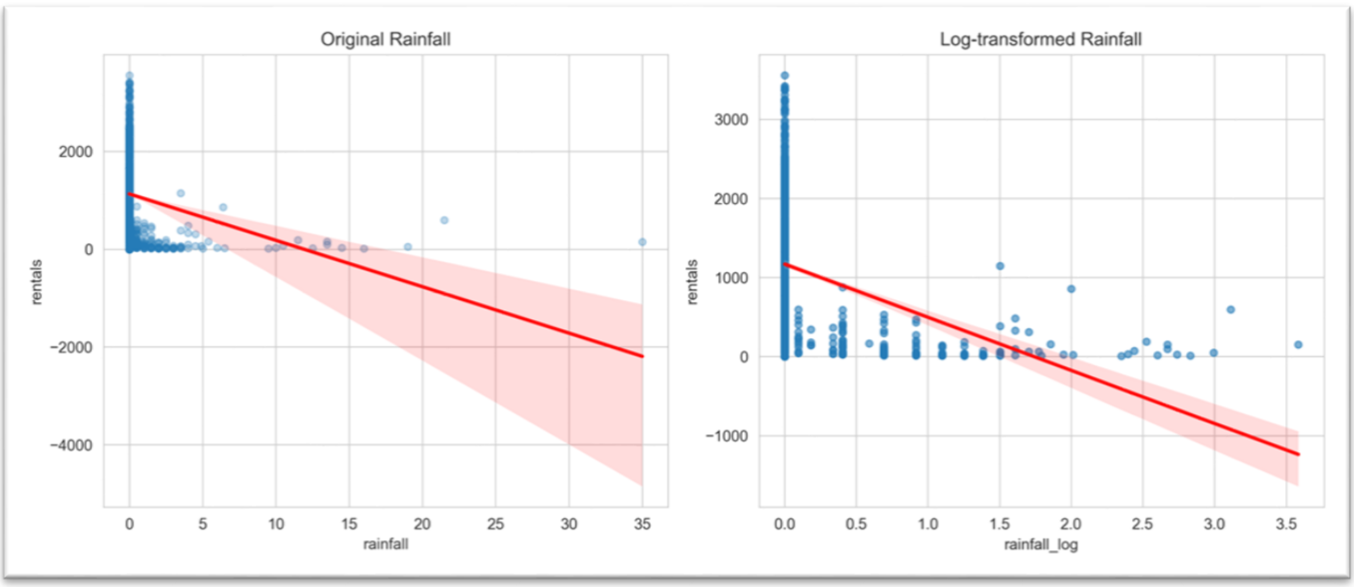

Figure 1: Rainfall vs. rentals, and log-transformed rainfall vs. rentals

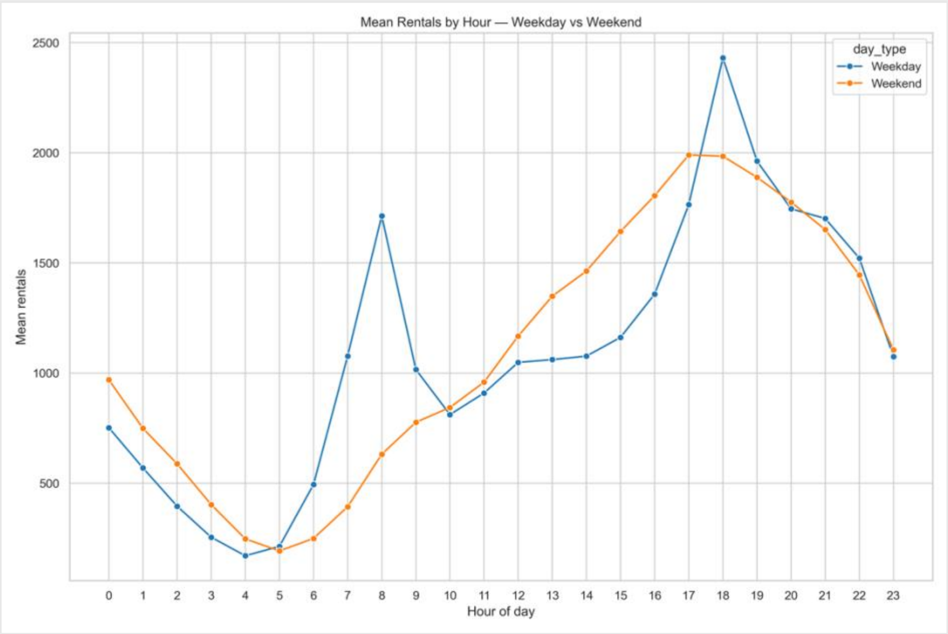

Figure 2: Hourly bike rental profiles by day type, showing weekday commuting peaks

Figure 3: Demand heatmap by day and hour, with highest rentals occuring on weekday evenings

Seoul's bike-sharing rental demand exhibits distinct temporal patterns that vary systematically by day type, specifically weekdays and weekends. In Figure 2, sharp bimodal peaks at 08:00 and 18:00 on weekdays align with typical commuting hours, consistent with utilitarian transport behaviour, while lower midday demand suggests limited usage during work hours. Meanwhile, a flatter distribution on weekends, with peak demand concentrated between 18:00 to 20:00, indicates leisure-oriented cycling, without the constraints of work schedules.

Similar patterns are seen in the mean rentals heatmap in Figure 3, where the highest aggregate demand occurs on weekday evenings, particularly Monday, Wednesday and Friday from 17:00 to 19:00. In contrast, weekend mornings (07:00 to 09:00) show lower activity compared to their weekday counterparts.

These temporal patterns mirror findings from Kinoshita, Bando and Sayama (2024), who demonstrated that bike-sharing systems primarily serve dual functions: commuter infrastructure on weekday mornings and evenings, and recreational amenities throughout the day on weekends. The systematic behavioural differences between weekdays and weekends observed here justify the inclusion of '**is_weekend**' as a binary predictor in subsequent regression analysis, as baseline demand levels differ fundamentally between these day types.

EDA – Weather-Rental Relationships, Correlation Matrix and Multicollinearity

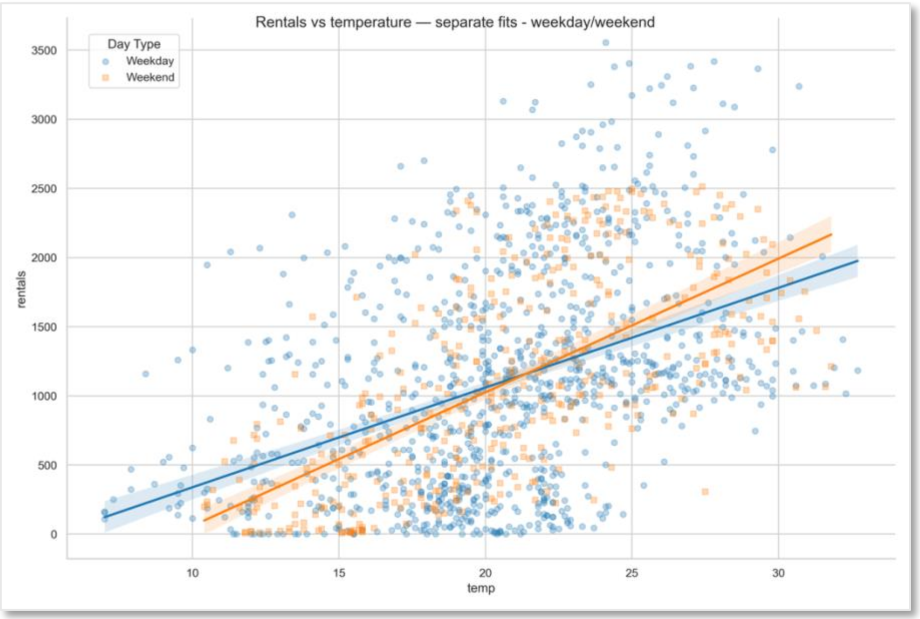

Figure 4: Temperature vs rentals by day type scatterplot, showing converging lines of best fit

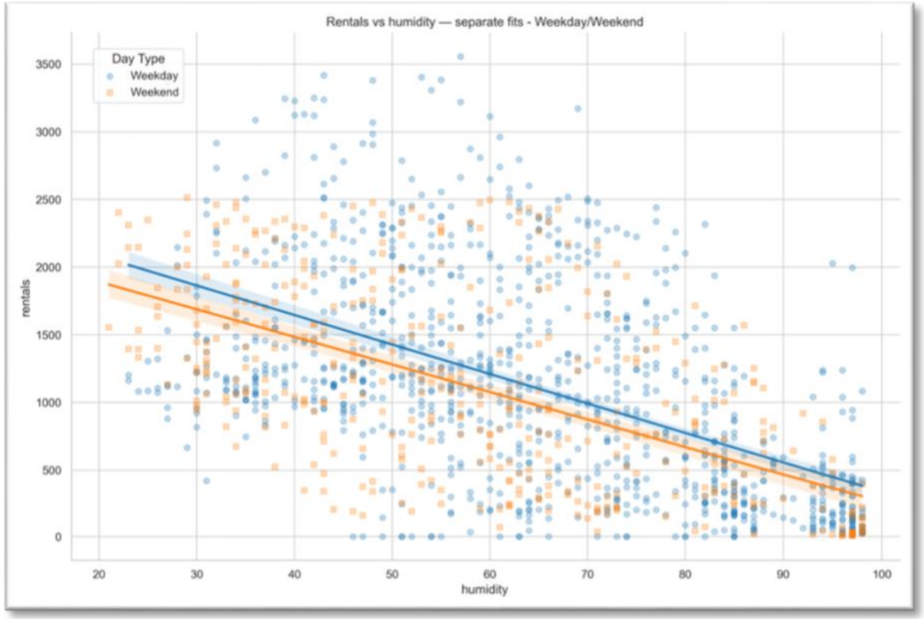

Figure 5: Humidity vs rentals by day type scatterplot, showing parallel lines of best fit

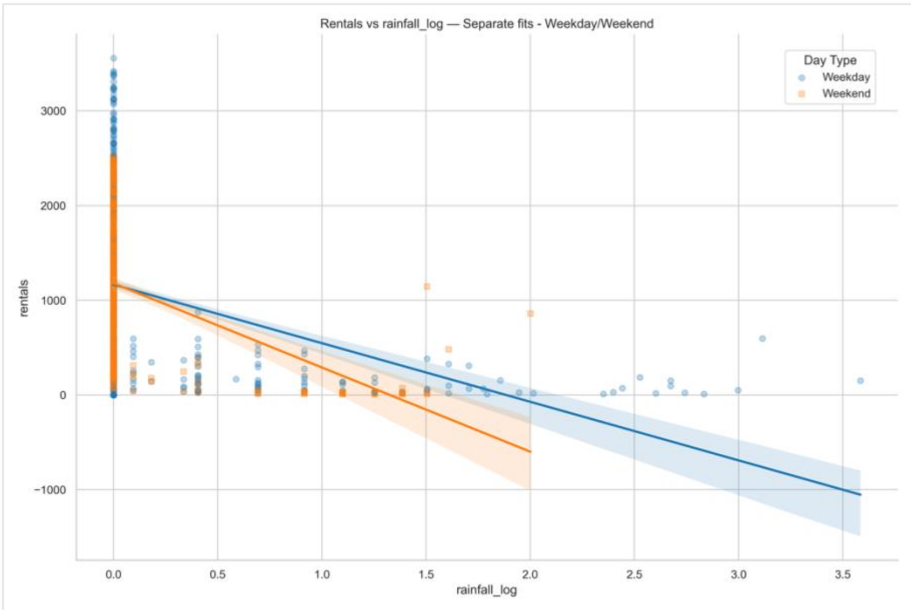

Figure 6: Rainfall vs rentals by day type scatterplot, showing diverging lines of best fit

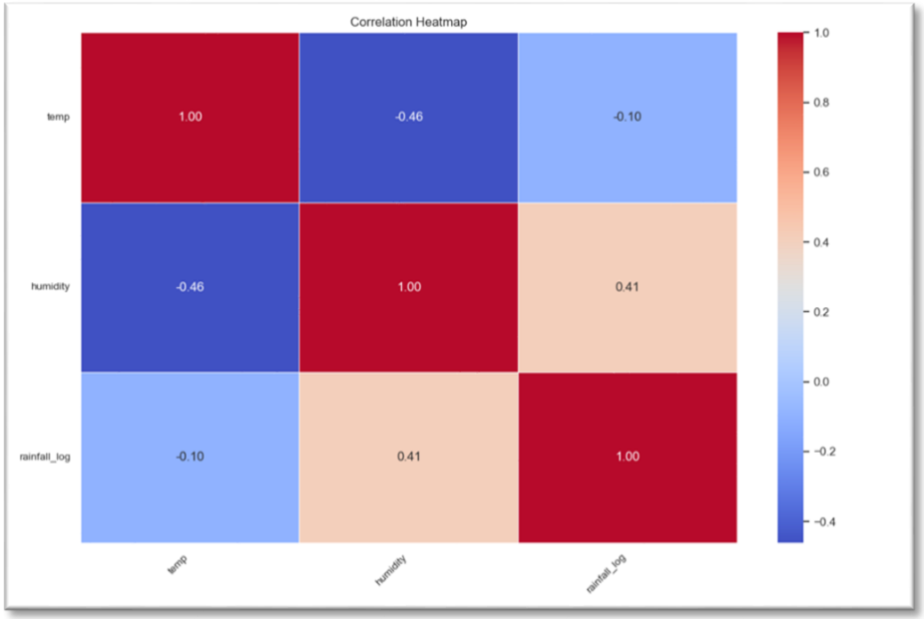

Figure 7: Correlation matrix between temperature, humidity and log-transformed rainfall

EDA – Weather-Rental Relationships

Temperature emerges as the strongest weather predictor of bike rental demand, showing a positive correlation across both weekdays and weekends. As illustrated in Figure 4, each 1°C increase in temperature corresponds to approximately 80 additional rentals per hour. Both groups converge at ~22°C, suggesting a shared optimal cycling temperature. The converging slopes between day types (weekdays = 72.1 rentals/°C, weekends = 96.6 rentals/°C) reveal that weekend demand increases more steeply as conditions warm. These patterns indicate that weekend leisure cyclists are more selective about higher temperatures and weather comfort, while weekday commuters maintain more consistent demand across temperature ranges.

Humidity trends exhibit a moderate negative correlation with rentals, though considerably weaker than the association with temperature. Figure 5 reveals that humidity over 70% notably deters cycling, with each 1% increase corresponding to around 22 fewer weekday rentals and 20 fewer weekend rentals. The parallel slopes indicate that humidity's deterring effect occurs uniformly across both utilitarian and recreational contexts. Physical discomfort from humid conditions, particularly when paired with elevated temperatures during Seoul's humid summer months, appears to influence users' decisions regardless of trip purpose and day type.

Figure 6 demonstrates how each 1-unit increase in log-transformed rainfall (equivalent to 1.72 mm of actual rainfall) corresponds to 619 fewer weekday rentals and 890 fewer weekend rentals. The diverging slopes and different x-intercepts reveal distinct behavioural differences between day types. Weekday commuters demonstrate greater rainfall resilience, maintaining positive demand until log-transformed rainfall exceeds approximately 1.85 units (≈ 5.36 mm), with narrower confidence bands indicating more predictable behaviour under adverse conditions. Meanwhile, weekend recreational users show heightened rain sensitivity, with steeper demand reduction and reaching zero demand at lower rainfall levels. This suggests that trip necessity, especially the importance of reaching work or school, forces weekday users to tolerate rainfall levels that would discourage discretionary leisure cycling.

EDA – Correlation Matrix and Multicollinearity

Before proceeding to multiple regression hypothesis testing, understanding the correlation structure among predictors is essential to identifying potential multicollinearity issues that could reverse regression coefficient signs, inflate standard errors and destabilise coefficient estimates. Figure 7's correlation matrix reveals that all bivariate correlations remain modest, with the highest absolute value being between temperature and humidity ($r = -0.46$) – below the conventional threshold of $|r| = 0.70$ signalling high multicollinearity.

To more rigorously assess multicollinearity for multiple regression modelling, Variance Inflation Factor (VIF) scores were calculated. VIF quantifies how much a predictor's variance is inflated due to correlations with other predictors. All three weather variables have low VIF scores: temperature (1.28), humidity (1.53) and log-transformed rainfall (1.22) – far below the VIF > 4 threshold for concern (Pennsylvania State University, 2018). This confirms that including all variables will yield reliable regression and hypothesis testing results.

With exploratory patterns established and multicollinearity ruled out, we now proceed to formal hypothesis testing through multiple regression to quantify the independent effects of weather conditions and day type on bike rental demand.

Hypothesis Testing

To test whether the effect of weather conditions on bike rental demand differs between weekdays and weekends, we use multiple regression with interaction terms. Interaction terms model the behaviour of one variable when it is affected by the value of another variable, such as weather with day type (Frost, 2017).

We specify the following hypotheses:

- H_0: **weather variables affect bike rental demand equally across weekdays and weekends.**
    - Mathematically, $\beta_5 = \beta_6 = \beta_7 = 0$ (see model below).
- H_1: **at least one weather variable exhibits a significantly different effect on bike rental demand between weekdays and weekends.**
    - Mathematically, $(\beta_5 \neq 0)$ OR $(\beta_6 \neq 0)$ OR $(\beta_7 \neq 0)$.

The multiple regression model used is:

$$\begin{aligned} rentals = \ & \beta_0 + \beta_1(temp) + \beta_2(rainfall\_log) + \beta_3(humidity) + \beta_4(is\_weekend) \\ & + \beta_5(temp \times is\_weekend) \\ & + \beta_6(rainfall\_\log \times is\_weekend) + \beta_7(\text{humidity} \times is\_weekend) + \varepsilon \end{aligned}$$

Where β_0 represents the intercept, or the baseline weekday demand under zero weather conditions, $\beta_1 - \beta_3$ captures baseline weekday weather effects, β_4 represents the weekend intercept change, $\beta_5 - \beta_7$ quantifies weather sensitivity and change on weekends, and ε represents the error term.

Using this model provides us with interaction terms, allowing us to 'toggle' the weekend effect. This yields distinct equations for each day type:

- Weekdays (**is_weekend = 0**):
$$rentals = \beta_0 + \beta_1(temp) + \beta_2(rainfall\_log) + \beta_3(humidity) + \varepsilon$$
- Weekends (**is_weekend = 1**):
$$rentals = (\beta_0 + \beta_4) + (\beta_1 + \beta_5)(temp) + (\beta_2 + \beta_6)(rainfall\_log) + (\beta_3 + \beta_7)(humidity) + \varepsilon$$

The weekend equation shows a shifted intercept of $(\beta_0 + \beta_4)$ showing a different baseline demand, while modified weather coefficients capture the combined change in rentals observed in exploratory analysis.

The model was estimated using Ordinary Least Squares (OLS) regression from Python's statsmodels library (see Appendix 1). Interaction terms (β_5, β_6, β_7) were created by multiplying the **is_weekend** indicator with all three weather predictors, allowing the model to estimate weather effects across day types.

Variable	Coefficient	Std. Error	t Stat	P-value	Lower 95%	Upper 95%
β_0 (intercept)	1191.9283	144.474	8.250	0.000	908.527	1475.330
β_1 (temp)	43.9112	4.541	9.670	0.000	35.004	52.819
β_2 (rainfall_log)	-306.3157	52.457	-5.839	0.000	-409.216	-203.415
β_3 (humidity)	-14.8415	1.172	-12.663	0.000	-17.141	-12.542
β_4 (is_weekend)	-918.0508	265.629	-3.456	0.001	-1439.113	-396.989
β_5 (temp x is_weekend)	25.3968	8.522	2.980	0.003	8.680	42.113
β_6 (rainfall_log x is_weekend)	46.0248	123.175	0.374	0.709	-195.596	287.646
β_7 (humidity x is_weekend)	4.8203	2.179	2.212	0.027	0.546	9.094

(**n** = 1,447; **multiple R** = 0.627; **R^2** = 0.393; **adjusted R^2** = 0.390; **SE** = 603.9; **significance F** = 6.31e-151)

Table 1: Multiple Regression Results for Bike Rental Demand with Weather and Day Type Interactions (see Appendix 1)

Statistical significance is established through t-statistics, which measure how many standard errors each coefficient deviates from zero. The corresponding p-values indicate the probability of observing a coefficient this extreme if the true effect were zero. Using a conventional significance level of $\alpha = 0.05$, coefficients with $p < 0.05$ are considered statistically significant. The F-statistic tests overall model significance; the p-value of the F-statistic is 6.31e-151, significantly less than 0.05, deeming our overall model statistically significant.

The weather interaction terms β_5 (temp x is_weekend) and β_7 (humidity x is_weekend) demonstrate statistical significance ($p = 0.003$ and $p = 0.027$, respectively). Temperature and humidity show significantly different effects between weekdays and weekends. Surprisingly, the rainfall interaction term β_6 (rainfall_log x is_weekend) is not statistically significant ($p = 0.709$), despite the line of best fit divergence observed in Figure 6. This suggests that while rainfall slopes appeared different between day types in EDA, statistical significance is weak due to the large standard error of 123.175.

- Temperature ($\beta_5 = 25.3968$, p-value = 0.003)
- Rainfall ($\beta_6 = 46.0248$, p-value = 0.709, *not significant*)
- Humidity ($\beta_7 = 4.8203$, p-value = 0.027)

Hence, we **reject our null hypothesis (H_0)** that weather variables affect bike rental demand equally across weekdays and **weekends** and conclude that **at least one weather variable exhibits significantly different effects between weekdays and weekends.**

Table 1 also reveals actionable information for Seoul's bike-sharing operations. On weekdays, each time the temperature increases by 1°C, approximately 44 more bikes are rented. Meanwhile, humidity exhibits a negative effect on bike-sharing: each time the humidity increases by 1%, about 15 fewer bikes are rented. Interaction terms allow us to model weekend weather effects on baseline weekday rental demand. Each time the temperature increases by 1°C on the weekend, 44 + 25 = 69 more bikes are rented. Conversely, humidity's effect is approximately 5 rentals/% less negative on weekends; each time humidity increases by 1% on the weekend, approximately 10 fewer bikes are rented (-14.8 + 4.8 = -10).

Using this information, bike-sharing operators can consider different business strategies. Operators could implement surge pricing on warm, less humid weekends, increasing per-minute rental rates to capitalise on higher predicted demand. Targeted resource allocation is another option, where more bikes are deployed to docking stations in anticipation of higher weekend demand during warm, less humid weather. On the other hand, operators may schedule fleet maintenance during cooler, more humid weekdays where demand is lower.

In summary, temperature has a stronger positive effect on bike rentals on weekends compared to weekdays (+25 rentals/°C), while humidity's deterrent effect is reduced on weekends (-10 vs. -15 rentals per 1%). Rainfall effects remain statistically insignificant across day types. These different weather sensitivities provide actionable insights for demand forecasting and operational planning in Seoul's bike-sharing system.

Limitations

Several data and methodological limitations should be considered when interpreting findings.

The two-month observational period, May and June 2018, limits the analysis to early summer behaviour, restricting generalisability to other seasons. While relationships between weather variables, day type and demand likely persist year-round, other conditions like winter weather introduce behaviours not observable in our current data. Freezing temperatures, snowfall, and icy roads may create non-linear effects or shift day-type interactions. To identify these patterns, an analysis of at least a full calendar year must be conducted to understand seasonal differences.

Beyond temporal constraints, geographic aggregation introduces additional limitations. Weather measurements represent city-wide conditions, likely sourced from a single meteorological station, hiding neighbourhood- and docking station-level differences in Seoul's diverse urban landscape. Docking stations near the Han River may experience cooler temperatures and higher humidity than those in the city centre. Beyond climate, station-level characteristics, such as subway connectivity, surrounding commercial activity, and nearby bike lanes, influence demand levels in ways that city-wide models cannot capture.

Similarly, hourly rental aggregation removes minute-by-minute data granularity; for example, a sudden ten-minute thunderstorm may cause sharp demand drops mid-hour, but this effect is averaged with pre-storm and post-storm rentals in the hourly total. Having higher-resolution, more granular data at the minute and station level would enable more precise spatial and temporal analysis of weather effects and day type interactions.

From the OLS model's R^2 value, 39.3% of observed rental variance can be explained using inputs in Table 1; this leaves 60.7% attributable to factors beyond weather and day type (Fernando, 2024). This is consistent with the model's standard error (SE) of 603.9, meaning typical predictions deviate by approximately ±604 rentals. While this represents substantial variation, it reflects the unpredictability of human decision-making and unmeasured contextual factors.

Several unmeasured variables, unavailable in our dataset, can contribute:

- **Air quality:** Seoul experiences high $PM_{2.5}$ pollution, frequently ranking among the most polluted global cities (IQAir, 2025). Official advice recommends residents avoid outdoor activity like cycling.
- **Special events:** concerts, sports games, protests, and national holidays create demand anomalies not captured by routine temporal patterns.
- **Supply constraints:** station-level bike availability limits realised demand; regardless of good weather conditions, users cannot rent from empty docking stations.
- **Individual preferences:** the model assumes uniform weather sensitivity, but preferences vary on a personal basis. Some cyclists brave any weather, while others cancel trips at the first raindrop.

Including these factors through richer datasets, integrating air quality monitors, event calendars, and real-time station inventory could improve predictive accuracy and business relevance.

Results reflect Seoul's specific context in May-June 2018 and may not generalise broadly. Geographic factors, like high urban density, an extensive subway system, and urban sprawl, shape cycling patterns distinct from flatter, car-centric Western cities (Jun et al., 2013). In addition, generalising this study's findings to modern users may also face challenges: the 2018 user base may differ from today's post-pandemic environment, where remote work and changed commuting patterns could alter weather and day type sensitivity. The bike-sharing system's age and user profiles also matter: early adopters in 2018 may exhibit different weather tolerance and recreational patterns than mainstream users in 2026 (Seoul Metropolitan Government, 2016). Caution is advised when applying these findings to inform policies in different geographic, temporal, or cultural contexts.

Discussion and Conclusions

This study investigated whether the impact of selected weather conditions on bike rental demand varies between weekdays and weekends in Seoul's bike-sharing system. The null hypothesis – that weather variables affect bike rental demand equally across weekdays and weekends – was rejected. Analysis revealed that temperature and humidity reveal significantly different effects across day types, while rainfall's effect remains statistically uniform across the week. These findings provide evidence that trip purpose, whether commuting or recreational, moderates weather sensitivity, with implications for operational forecasting and planning.

Three key findings emerged from the analysis. First, temperature exerts the strongest positive influence on demand; each 1°C increase corresponds to approximately 44 rentals on weekdays and 69 rentals on weekends ($\beta_5 = 25.3968$, $p = 0.003$). Second, humidity has a moderate negative effect on demand, which is less pronounced on weekends – each percentage point reduces demand by 15 rentals on weekdays versus 10 on weekends ($\beta_7 = 4.8203$, $p = 0.027$). Third, rainfall's log-transformed deterrent effect operates uniformly across day types ($\beta_6 = 46.0248$, $p = 0.709$), contrary to exploratory analysis illustrating a divergence in lines of best fit between weekday and weekend plots.

These findings reveal behavioural differences in bike-sharing users driven by trip purpose. The stronger temperature sensitivity on weekends suggests that recreational cyclists optimise for comfort, prioritising warmer conditions for leisure rides. In contrast, weekday commuters maintain more consistent demand across a range of temperatures, reflecting the need to reach work or school regardless of the weather. Similarly, humidity's weaker weekend effect indicates that weekend users are slightly more likely to tolerate humid conditions, as trips are discretionary, flexible in timing, and less constrained by work requirements. On weekdays, higher humidity more strongly suppresses demand, likely because discomfort and perceived inconvenience would increase the attractiveness of alternative transport modes like buses or subways.

The absence of a statistically significant difference in rainfall effect across day types should also be discussed. While exploratory analysis suggested that weekend users exhibit greater rain sensitivity, hypothesis testing revealed that this difference lacks statistical significance ($p = 0.709$). This finding suggests that rain may act as a universal deterrent; both commuters and recreational users avoid cycling in wet conditions at similar rates. The visual divergence in lines of best fit, observed in EDA, is likely explained by high standard errors (seen in Table 1) rather than a genuine behavioural difference, highlighting the importance of rigorous hypothesis testing to validate patterns.

These findings enable evidence-based, operational strategies for Seoul's bike-sharing system. For example, dynamic pricing schemes could capitalise on optimised weekend weather conditions, increasing per-minute rates during warm, low-humidity weekends when demand peaks. In addition, resource allocation can be improved by deploying additional bikes to high-traffic docking stations during favourable weekend weather, when recreational demand surges beyond baseline levels. Finally, maintenance scheduling should prioritise cool, humid weekdays when demand is lowest, minimising service disruptions during peak periods.

Several topics for future research emerge from this study's limitations. Extending the analysis to a full calendar year would reveal whether weather and day-type interactions vary seasonally, and model how winter cycling under freezing temperatures and snow may exhibit different patterns than the summer data analysed here. Neighbourhood- and station-level models could provide more granular spatial details on whether weather sensitivity differs between riverside locations, urban centres, and near transit links. On the other hand, incorporating unmeasured variables – particularly air quality, special events, and real-time bike availability – could improve predictive accuracy and address the remaining 60.7% unexplained variance in our model.

This study demonstrates that the weather's influence on bike-sharing demand is not uniform; the predominant trip purpose, depending on the day of the week, fundamentally shapes how users respond to their environment. By visualising and quantifying these different effects, the analysis provides actionable insights and recommendations for operational planning while contributing to transportation behaviour theory by demonstrating that weather sensitivity is conditional on trip purpose rather than uniform across time. As bike-sharing systems continue to expand globally, understanding the relationships between weather, temporal patterns, and human behaviour will be key to optimising micromobility infrastructure.

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Appendix 1: Python OLS Regression Results

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                            OLS Regression Results
==============================================================================
Dep. Variable:          Q('rentals')   R-squared:                       0.393
Model:                           OLS   Adj. R-squared:                  0.390
Method:                Least Squares   F-statistic:                     132.9
Date:               Sun, 04 Jan 2026   Prob (F-statistic):          6.31e-151
Time:                       16:10:46   Log-Likelihood:                -11315.
No. Observations:               1447   AIC:                         2.265e+04
Df Residuals:                   1439   BIC:                         2.269e+04
Df Model:                          7
Covariance Type:           nonrobust
================================================================================================
                                  coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------------------------
Intercept                    1191.9283    144.474      8.250      0.000     908.527    1475.330
Q('temp')                      43.9112      4.541      9.670      0.000      35.004      52.819
Q('rainfall_log')            -306.3157     52.457     -5.839      0.000    -409.216    -203.415
Q('humidity')                 -14.8415      1.172    -12.663      0.000     -17.141     -12.542
is_weekend                   -918.0508    265.629     -3.456      0.001   -1439.113    -396.989
Q('temp'):is_weekend           25.3968      8.522      2.980      0.003       8.680      42.113
Q('rainfall_log'):is_weekend   46.0248    123.175      0.374      0.709    -195.596     287.646
Q('humidity'):is_weekend        4.8203      2.179      2.212      0.027       0.546       9.094
==============================================================================
Omnibus:                      86.452   Durbin-Watson:                   0.407
Prob(Omnibus):                 0.000   Jarque-Bera (JB):              101.607
Skew:                          0.649   Prob(JB):                     8.64e-23
Kurtosis:                      2.994   Cond. No.                     1.27e+03
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Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[2] The condition number is large, 1.27e+03. This might indicate that there are
strong multicollinearity or other numerical problems.

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Standard Error (SE): 603.914

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Ordinary Least Squares (OLS) regression output generated using Python's statsmodels library.

The table reports estimated coefficients, standard errors, t-statistics, p-values, and 95% confidence intervals for the multiple regression model with weather and day-type interaction terms. Results summarised and interpreted in the Hypothesis Testing section.